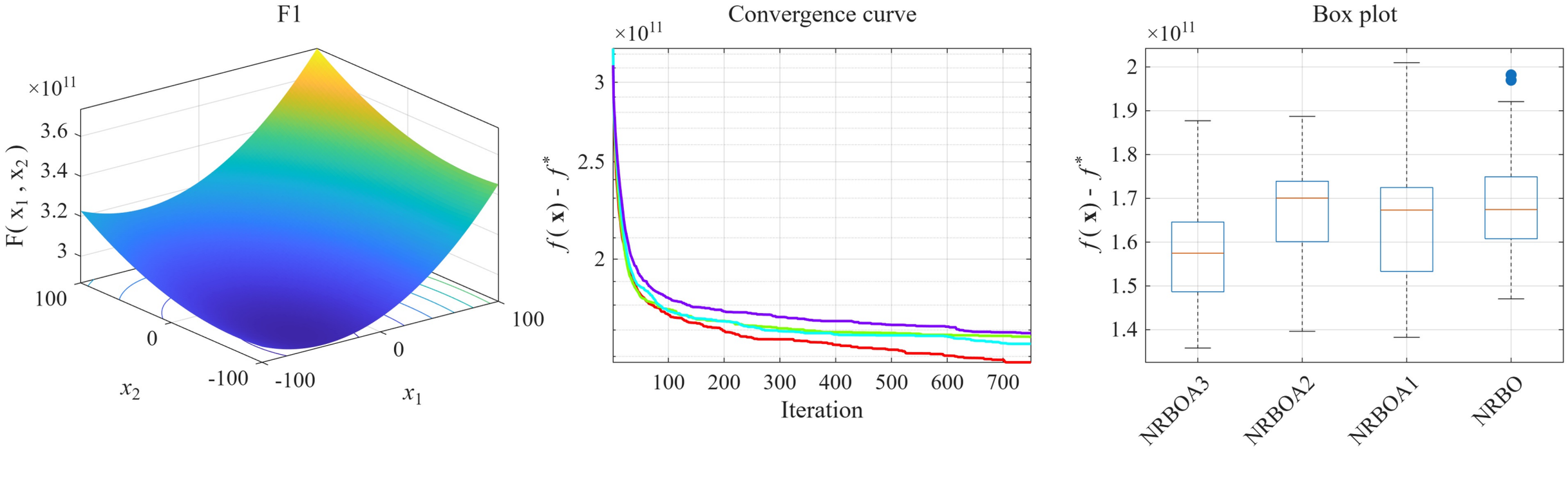

F1
×10^11
3.6
3.4
3.2
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^11
3
2.5
2
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^11
2
1.9
1.8
1.7
1.6
1.5
1.4
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO